322613 (22)

BE (6th Semester)

Examination, April - May, 2016

(Old Scheme)

UNIX and Shell Programming

Time Allowed : 3 hours *Maximum Marks* : 80

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Minimum Pass Marks : 28

Note : (*i*) Part (*a*) is compulsory and attempt any two parts from (*b*), (*c*) and (*d*) of each question.

(*ii*) The figures in the right-hand margin indicate marks.

1. (*a*) How can one should change the password of current user logged in Unix operating system through command line? 2

(*b*) Discuss different modes of Vi-Editor and specify the basic commands for the same. 7

(*c*) What is the role of KERNEL in Unix operating system? Explain various features of KERNEL. 7

(*d*) Discuss the various approaches of setting file permissions in UNIX. 7

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2. (*a*) Write command to list all the links from a directory. 2

(*b*) Explain the process of creating environment variables. Determine what the built-in shell variables are for your account. Which ones are exportable? 7

(*c*) What is shell programming? Write a shell script to create a menu which displays the list of files, current users and contents of particular file and process status of the system based on the user choice. 7

(*d*) Explain all the decision-making statements in detail. 7

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3. (*a*) Explain with example different types of files supported in UNIX. 2

(*b*) Explain the mechanism of process creation and role of system calls. 7

(*c*) Write a short note on file mounting and unmounting in Unix. 7

(*d*) Discuss the advantages and disadvantages of Buffer cache. 7

4. (*a*) Differentiate hard link with symbolic link. 2

(*b*) What is Zombie process in UNIX? How do you find Zombie process in UNIX? 7

(*c*) Write an algorithm for assigning inode (iget). 7

(Continued)

(*d*) How will you run a process in background? How will you bring that into foreground and how will you kill that process? 7

5. (*a*) Specify the terminal command used for installing package on Ubuntu and Fedora Linux. 2

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(*b*) Write an algorithm for wakeup system call. 7

(*c*) Describe the process of Installing Ubuntu or Fedora installation in dual boot mode. 7

(*d*) What is process? What are the different states of process in Unix? Explain using transition state diagram. 7

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